Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Innovating Product Development with 3D Printing and Additive Manufacturing

TGS Ref No: TGS-2022017597

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 8.1

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
7.0	1 June 2025	Legacy release — v7 trainer deck and schedule.	Dr Alfred Ang
8.0	18 August 2026	Full regeneration: single-source rebuild aligned to the v8.0 slide deck (150 slides), 16 hands-on labs in per-lab folders, enriched AM content (NIST/ISO-ASTM process families, materials, case studies) and the WA (SAQ) + PP assessment.	Dr Alfred Ang
8.1	22 August 2026	Aligned the 150-slide deck and 16-lab schedule to the supplied diagnostic STL/Fusion/Cura assets; added evidence-based calibration, support, retraction, cooling, wall, optimisation and finishing tasks.	Dr Alfred Ang

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Course Information

Course Title	Innovating Product Development with 3D Printing and Additive Manufacturing
WSQ Course Reference	TGS-2022017597
Skills Framework TSC	Additive Manufacturing (PRE-OPR-4012-1.1)
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 training hours per day (16 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on labs with Fusion 360, Ultimaker Cura and FDM 3D printers
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Analyze the AM and evaluate the use of reverse engineering using 3D scanning
- LO2: Analyze the issues and determine the requirements for AM processes
- LO3: Apply AM optimization techniques and assess the impact of process improvement

TSC Abilities and Knowledge

This course delivers the following TSC abilities and knowledge factors (Additive Manufacturing, PRE-OPR-4012-1.1):

- A1: Analyse and determine material considerations of components to be manufactured
- A2: Analyse process-related issues for using DMLS, SLS, SLM, powder bed and inkjet head, and/or 3D printing for AM
- A3: Determine processes and procedures for manufacturing metallic and non-metallic components using AM
- A4: Evaluate use of reverse engineering processes for components through 3D scanning and subsequent AM
- A5: Apply optimisation techniques to improve efficiency of AM processes and product quality
- A6: Assess the impact of manufacturing process improvements on business operations
- K1: AM characteristics, techniques and systems
- K2: Requirements of different AM processes
- K3: Factors to be considered when selecting AM solutions
- K4: Impact of AM on manufacturing operations
- K5: Methods for reviewing AM performance
- K6: Benefits and trade-offs of AM
- K7: Methods of ensuring long-term sustainability of AM

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Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — scenario-based practical tasks, 90 minutes, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 2 from 4:00 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Additive Manufacturing, 3D Scanning and Design for AM

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, trainer and learner introductions, course outline, learning outcomes, ground rules and mandatory digital attendance (AM)
10:00–11:00	60 min	**Topic 1 — Overview of Additive Manufacturing: definition, AM vs subtractive, history, the 3D printing workflow, the 7 AM process families, filament/resin/powder printing, DMLS/SLS/SLM (concepts + demo) (Slides 18 –72)**
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	Topic 1 — Design for AM in Fusion 360. Hands-on: Lab 1: Extrude — Personalised Keychain Token (Slide 54); Lab 2: Sweep — Giant Paper Clip Bookmark (Slide 57); Lab 3: Loft — Mini Plant Pot with Custom Rails (Slide 60)
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Topic 1 — Materials, benefits & trade-offs, case studies and AM selection factors (concepts). Digital attendance (PM). Hands-on: Lab 4: Revolve — Funnel from a 2D Profile (Slide 63); Lab 5: Tolerance — Container and Lid (0.2 mm Fit) (Slide 66)
15:30–15:45	15 min	Tea break
15:45–17:45	120 min	Topic 1 — 3D scanning and reverse engineering: scanning apps, mesh-to-CAD, case studies (concepts). Hands-on: Lab 6: 3D Scan and Print a Real Object (Slide 69)

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Time	Duration	Topic / Activity
17:45–18:30	45 min	**Topic 2 begins — Where AM issues come from; the slicer. Hands-on: Lab 7: Install Cura and Slice Your First Model (Slide 94). Day 1 recap and Q&A**

Total training time: 480 minutes (8 hours).

Day 2 — Process Issues, Process Improvement and Assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	**Topic 2 — Printer anatomy, bed adhesion, raft/brim/skirt and supports (concepts + demo) (Slides 73–106). Hands-on: Lab 8: Bed Adhesion — Skirt vs Brim vs Raft (Slide 97)**
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	Topic 2 — Extrusion issues, retraction, oozing, temperature, jams, filament types and hybrid metal/non-metal manufacture. Hands-on: Lab 9: Support Structures for Overhangs (Slide 100); Lab 10: Filament Diagnostics — Diameter, Temperature and Flow (Slide 103)
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Topic 3 — Process Improvement: infill, surface quality, textures, supports, post-processing, business impact and sustainability (concepts) (Slides 107–140). Digital attendance (PM). Hands-on: Lab 11: Perimeters and Infill — Strength vs Speed (Slide 124); Lab 12: Infill Patterns — Zigzag vs Grid Strength Test (Slide 126); Lab 13: Layer Height and Top Thickness — Surface Quality (Slide 129); Lab 14: Ironing — Glass-Smooth Top Surfaces (Slide 131); Lab 15: Fuzzy Skin — Textured Walls on a Vase (Slide 134); Lab 16: Post-Processing — Support Removal and Sanding (Slide 137)
15:30–15:45	15 min	Tea break

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Time	Duration	Topic / Activity
15:45–16:00	15 min	**Revision, course feedback, TRAQOM survey, Assessment digital attendance and Briefing for Assessment**
16:00–17:00	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book**
17:00–18:30	90 min	**Practical Performance (PP) — scenario-based practical tasks, 90 minutes, open book**

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to the TSC)

Topic	Learning Outcome	Labs
Topic 01: Overview of Additive Manufacturing and 3D Scanning	LO1	Lab 1, Lab 2, Lab 3, Lab 4, Lab 5, Lab 6
Topic 02: Process Related Issues for AM	LO2	Lab 7, Lab 8, Lab 9, Lab 10
Topic 03: Process Improvement for AM	LO3	Lab 11, Lab 12, Lab 13, Lab 14, Lab 15, Lab 16

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